# 五 多元函数微分学

## 一、选择题

**1.(C).** 在 $F(xz,yz)=0$ 两边分别对 $x,y$ 求偏导,得

$$F'_1\cdot\left(z+x\frac{\partial z}{\partial x}\right)+yF'_2\cdot\frac{\partial z}{\partial x}=0,xF'_1\cdot\frac{\partial z}{\partial y}+F'_2\cdot\left(z+y\frac{\partial z}{\partial y}\right)=0,$$

解得 $\frac{\partial z}{\partial x}=-\frac{zF'_1}{xF'_1+yF'_2},\frac{\partial z}{\partial y}=-\frac{zF'_2}{xF'_1+yF'_2}$,则 $x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=-z$. 选(C).

**2.(C).** 因为 $\lim\limits_{x\to0}\frac{f(x,0)-f(0,0)}{x}=\lim\limits_{x\to0}\frac{\sin\sqrt{x^2}}{x}=\lim\limits_{x\to0}\frac{\sqrt{x^2}}{x}=\lim\limits_{x\to0}\frac{|x|}{x}$ 不存在,所以 $f'_x(0,0)$ 不存在;由 $\lim\limits_{y\to0}\frac{f(0,y)-f(0,0)}{y}=\lim\limits_{y\to0}\frac{\sin\sqrt{y^4}}{y}=\lim\limits_{y\to0}\frac{\sqrt{y^4}}{y}=\lim\limits_{y\to0}y=0$,得 $f'_y(0,0)=0$. 选(C).

**3.(C).** 因为 $\lim\limits_{\substack{x\to0\\y\to0}}f(x,y)=0=f(0,0)$,所以 $f(x,y)$ 在(0,0)处连续;

因为 $\lim\limits_{x\to0}\frac{f(x,0)-f(0,0)}{x}=\lim\limits_{x\to0}\frac{x^2\sin\frac{1}{x^2}}{x}=\lim\limits_{x\to0}x\sin\frac{1}{x^2}=0$,所以 $f'_x(0,0)=0$,根据对称性,$f'_y(0,0)=0$,即 $f(x,y)$ 在(0,0)处可偏导;

由 $\lim\limits_{\rho\to0}\frac{f(x,y)-f(0,0)-f'_x(0,0)x-f'_y(0,0)y}{\rho}=\lim\limits_{\rho\to0}\rho\sin\frac{1}{\rho^2}=0(\rho=\sqrt{x^2+y^2})$,得 $f(x,y)$ 在(0,0)处可微;

当 $(x,y)\neq(0,0)$ 时,$f'_x(x,y)=2x\sin\frac{1}{x^2+y^2}-\frac{2x}{x^2+y^2}\cos\frac{1}{x^2+y^2}$,则

$$f'_x(x,y)=\begin{cases}2x\sin\dfrac{1}{x^2+y^2}-\dfrac{2x}{x^2+y^2}\cos\dfrac{1}{x^2+y^2}, & (x,y)\neq(0,0),\\ 0, & (x,y)=(0,0),\end{cases}$$

因为 $\lim\limits_{\substack{x\to0\\y\to0}}f'_x(x,y)=\lim\limits_{\substack{x\to0\\y\to0}}\left(2x\sin\frac{1}{x^2+y^2}-\frac{2x}{x^2+y^2}\cos\frac{1}{x^2+y^2}\right)$ 不存在,所以 $f'_x(x,y)$ 在点(0,0)处不连续,同理,$f'_y(x,y)$ 在点(0,0)处也不连续. 选(C).

**4.(C).** 因为若函数 $f(x,y)$ 一阶连续可偏导,则 $f(x,y)$ 一定可微,反之则不对,即若函数 $f(x,y)$ 偏导数不连续,则 $f(x,y)$ 不一定不可微. 选(C).

**5.(D).** 令 $\rho=\sqrt{(x-1)^2+y^2}$,由 $\lim\limits_{\substack{x\to1\\y\to0}}\frac{f(x,y)-3x-2y-1}{e^{(x-1)^2+y^2}-1}=-2$,得 $f(1,0)=4$.

再由 $\lim\limits_{\substack{x\to1\\y\to0}}\frac{f(x,y)-3x-2y-1}{e^{(x-1)^2+y^2}-1}=\lim\limits_{\substack{x\to1\\y\to0}}\frac{f(x,y)-3x-2y-1}{(x-1)^2+y^2}=-2$,得

$$f(x,y)-3x-2y-1=o(\rho)\text{ 或 }f(x,y)=3x+2y+1+o(\rho),$$

则 $\Delta z=f(x,y)-f(1,0)=3(x-1)+2(y-0)+o(\rho)$,由可微的定义得 $z=f(x,y)$ 在(1,0)处可微,且 $dz|_{(1,0)}=3dx+2dy$. 选(D).

**6.(C).** 令 $\rho=\sqrt{(x-1)^2+y^2}$,由 $\lim\limits_{\substack{x\to1\\y\to0}}\frac{f(x,y)-3x+y-2}{(x-1)^2+y^2}=-1$,得 $f(1,0)=5$ 且 $f(x,y)-3x+y-2=o(\rho)$,即 $f(x,y)=3x-y+2+o(\rho)$.

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$\Delta z=f(x,y)-f(1,0)=3(x-1)-(y-0)+o(\rho)$，则 $dz|_{(1,0)}=3dx-dy$，即

$$f'_x(1,0)=f'_1(1,0)=3,f'_y(1,0)=f'_2(1,0)=-1.$$

则 $\lim\limits_{h\to0}\dfrac{f(e^{2h},h)-f(e^{-h},-2h)}{h}$

$$=\lim_{h\to0}\frac{f(e^{2h},h)-f(1,h)}{h}+\lim_{h\to0}\frac{f(1,h)-f(1,-2h)}{h}+\lim_{h\to0}\frac{f(1,-2h)-f(e^{-h},-2h)}{h}$$

$$=\lim_{h\to0}f'_1(\xi_1,h)\cdot\frac{e^{2h}-1}{h}+3\lim_{h\to0}f'_2(1,\xi_2)+\lim_{h\to0}f'_1(\xi_3,-2h)\cdot\frac{1-e^{-h}}{h},$$

其中 $\xi_1$ 在 1 与 $e^{2h}$ 之间，$\xi_2$ 在 $h$ 与 $-2h$ 之间，$\xi_3$ 在 1 与 $e^{-h}$ 之间.

故 $\lim\limits_{h\to0}\dfrac{f(e^{2h},h)-f(e^{-h},-2h)}{h}=2f'_1(1,0)+3f'_2(1,0)+f'_1(1,0)=6$. 选(C).

## 二、填空题

**1. 解** 由 $z=xf(x+y)+g(x^y,x^2+y^2)$，得

$$\frac{\partial z}{\partial x}=f(x+y)+xf'(x+y)+yx^{y-1}g'_1(x^y,x^2+y^2)+2xg'_2(x^y,x^2+y^2),$$

$$\frac{\partial^2 z}{\partial x\partial y}=f'+xf''+x^{y-1}g'_1+yx^{y-1}\ln xg'_1+yx^{2y-1}\ln xg''_{11}+2y^2x^{y-1}g''_{12}+2x^{y+1}\ln xg''_{21}+4xyg''_{22}.$$

**2. 解** 在 $u(x,3x)=x$ 两边同时对 $x$ 求导，得 $u'_x(x,3x)+3u'_y(x,3x)=1$.

再对 $x$ 求导，得 $u''_{xx}(x,3x)+6u''_{xy}(x,3x)+9u''_{yy}(x,3x)=0$.

由 $\dfrac{\partial^2 u}{\partial x^2}=\dfrac{\partial^2 u}{\partial y^2}$，得 $10u''_{xx}(x,3x)+6u''_{xy}(x,3x)=0$.

$u'_x(x,3x)=x^3$ 两边对 $x$ 求导，得 $u''_{xx}(x,3x)+3u''_{xy}(x,3x)=3x^2$. 解得

$$u''_{xy}(x,3x)=\frac{5x^2}{4}.$$

**3. 解** 因为 $F'_x\neq0$，所以由 $F(x,y,z)=0$ 可确定 $x$ 为 $y,z$ 的函数.

$F(x,y,z)=0$ 两边对 $y$ 求偏导，得 $F'_x\cdot\dfrac{\partial x}{\partial y}+F'_y=0$，解得 $\dfrac{\partial x}{\partial y}=-\dfrac{F'_y}{F'_x}$.

因为 $F'_y\neq0$，所以由 $F(x,y,z)=0$，可确定 $y$ 为 $x,z$ 的函数.

$F(x,y,z)=0$ 两边对 $z$ 求偏导，得 $F'_y\cdot\dfrac{\partial y}{\partial z}+F'_z=0$，解得 $\dfrac{\partial y}{\partial z}=-\dfrac{F'_z}{F'_y}$.

因为 $F'_z\neq0$，所以由 $F(x,y,z)=0$，可确定 $z$ 为 $x,y$ 的函数.

$F(x,y,z)=0$ 两边对 $x$ 求偏导，得 $F'_x+F'_z\cdot\dfrac{\partial z}{\partial x}=0$，解得 $\dfrac{\partial z}{\partial x}=-\dfrac{F'_x}{F'_z}$.

故 $\dfrac{\partial x}{\partial y}\cdot\dfrac{\partial y}{\partial z}\cdot\dfrac{\partial z}{\partial x}=-1$.

**4. 解** $z=x^{2y}+y^{\sin^2x}=e^{2y\ln x}+e^{\sin^2x\ln y}$，

$\dfrac{\partial z}{\partial x}=2yx^{2y-1}+y^{\sin^2x}\sin 2x\cdot\ln y$，$\dfrac{\partial z}{\partial y}=2x^{2y}\ln x+y^{\sin^2x}\cdot\dfrac{\sin^2x}{y}$，则

$$dz=(2yx^{2y-1}+y^{\sin^2x}\sin 2x\cdot\ln y)dx+\left(2x^{2y}\ln x+y^{\sin^2x}\cdot\frac{\sin^2x}{y}\right)dy.$$

**5. 解** 当 $x=0,y=1$ 时，$z=2$. $e^{x+2y-z}=1+xy^{\frac{3}{2}}z$ 两边求微分，得

$$e^{x+2y-z}\cdot(dx+2dy-dz)=y^{\frac{3}{2}}zdx+\frac{3}{2}xy^{\frac{1}{2}}zdy+xy^{\frac{3}{2}}dz,$$

将 $x=0,y=1,z=2$ 代入，得 $dz|_{(0,1)}=-dx+2dy$.

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**6. 解** 由 $\lim\limits_{\substack{x\to 1\\ y\to -1}}\dfrac{f(x,y)-2x+y-3}{(x-1)^2+(y+1)^2}=2$，得 $f(1,-1)=6$，令 $\rho=\sqrt{(x-1)^2+(y+1)^2}$，

由 $\lim\limits_{\substack{x\to 1\\ y\to -1}}\dfrac{f(x,y)-2x+y-3}{(x-1)^2+(y+1)^2}=2$，得 $f(x,y)-2x+y-3=o(\rho)$，即

$$\Delta z=f(x,y)-f(1,-1)=2(x-1)-(y+1)+o(\rho),$$

由微分定义得 $\mathrm{d}z\mid_{(1,-1)}=2\mathrm{d}x-\mathrm{d}y$.

**7. 解** 在 $f(tx,ty)=t^3f(x,y)$ 两边同时对 $t$ 求导，得

$$xf_1'(tx,ty)+yf_2'(tx,ty)=3t^2f(x,y),$$

取 $t=1,x=1,y=2$，得 $f_1'(1,2)+2f_2'(1,2)=3f(1,2)$，故 $f(1,2)=3$.

**8. 解** 由 $\dfrac{\partial^2 z}{\partial y^2}=2$，得 $\dfrac{\partial z}{\partial y}=2y+\varphi(x)$，因为 $f_y'(x,0)=x$，所以 $\varphi(x)=x$，即 $\dfrac{\partial z}{\partial y}=2y+x$，进一步有 $z=y^2+xy+C$，因为 $f(x,0)=1$，所以 $C=1$，于是 $z=y^2+xy+1$.

## 三、解答题

**1. 解** $\dfrac{\partial z}{\partial x}=2x\mathrm{e}^{x^2+y^2}$，$\dfrac{\partial^2 z}{\partial x^2}=2\mathrm{e}^{x^2+y^2}+4x^2\mathrm{e}^{x^2+y^2}$，

$\dfrac{\partial z}{\partial y}=2y\mathrm{e}^{x^2+y^2}$，$\dfrac{\partial^2 z}{\partial y^2}=2\mathrm{e}^{x^2+y^2}+4y^2\mathrm{e}^{x^2+y^2}$.

则 $\dfrac{\partial^2 z}{\partial x^2}+\dfrac{\partial^2 z}{\partial y^2}=4[1+(x^2+y^2)]\mathrm{e}^{x^2+y^2}$.

**2. 解** $\dfrac{\mathrm{d}z}{\mathrm{d}t}=2\mathrm{e}^{2t}f_1'+\sin 2t\cdot f_2'$，

$$\begin{aligned}\frac{\mathrm{d}^2 z}{\mathrm{d}t^2}&=4\mathrm{e}^{2t}f_1'+2\mathrm{e}^{2t}(2\mathrm{e}^{2t}f_{11}''+\sin 2t\cdot f_{12}'')+2\cos 2t\cdot f_2'+\sin 2t\cdot(2\mathrm{e}^{2t}f_{21}''+\sin 2t\cdot f_{22}'')\\&=4\mathrm{e}^{2t}f_1'+4\mathrm{e}^{4t}f_{11}''+4\mathrm{e}^{2t}\sin 2t\cdot f_{12}''+2\cos 2t\cdot f_2'+\sin^2 2t\cdot f_{22}''.\end{aligned}$$

**3. 解** $\dfrac{\partial z}{\partial x}=2xf_1'+\cos x\cdot f_2'-y\sin x\cdot f_3'$，

$$\begin{aligned}\frac{\partial^2 z}{\partial x\partial y}&=2x(2yf_{11}''+\cos x\cdot f_{13}'')+\cos x\cdot(2yf_{21}''+\cos x\cdot f_{23}'')-\sin x\cdot f_3'-\\&\quad y\sin x\cdot(2yf_{31}''+\cos x\cdot f_{33}'')\\&=4xyf_{11}''+(2x\cos x-2y^2\sin x)\cdot f_{13}''+2y\cos x\cdot f_{21}''+\cos^2 x\cdot f_{23}''-\sin x\cdot f_3'-\\&\quad y\sin x\cos x\cdot f_{33}''.\end{aligned}$$

**4. 解** 由 $z=(x^2+y^2)^{\sec^2(x+y)}$，得 $z=\mathrm{e}^{\sec^2(x+y)\ln(x^2+y^2)}$，则

$$\frac{\partial z}{\partial x}=(x^2+y^2)^{\sec^2(x+y)}\left[2\sec^2(x+y)\tan(x+y)\ln(x^2+y^2)+\frac{2x}{x^2+y^2}\sec^2(x+y)\right],$$

$$\frac{\partial z}{\partial y}=(x^2+y^2)^{\sec^2(x+y)}\left[2\sec^2(x+y)\tan(x+y)\ln(x^2+y^2)+\frac{2y}{x^2+y^2}\sec^2(x+y)\right].$$

**5. 解** 方程组由五个变量三个方程构成，故确定了三个二元函数，其中 $x,y$ 为自变量，由 $u=f(x,y,z,t),g(y,z,t)=0,h(z,t)=0$，得

$$\begin{cases}\dfrac{\partial u}{\partial x}=f_1'+f_3'\dfrac{\partial z}{\partial x}+f_4'\dfrac{\partial t}{\partial x},\\[2mm]\dfrac{\partial g}{\partial z}\dfrac{\partial z}{\partial x}+\dfrac{\partial g}{\partial t}\dfrac{\partial t}{\partial x}=0,\\[2mm]\dfrac{\partial h}{\partial z}\dfrac{\partial z}{\partial x}+\dfrac{\partial h}{\partial t}\dfrac{\partial t}{\partial x}=0,\end{cases}\qquad \begin{vmatrix}\dfrac{\partial g}{\partial z}&\dfrac{\partial g}{\partial t}\\[2mm]\dfrac{\partial h}{\partial z}&\dfrac{\partial h}{\partial t}\end{vmatrix}=\frac{\partial g}{\partial z}\frac{\partial h}{\partial t}-\frac{\partial g}{\partial t}\frac{\partial h}{\partial z}\neq 0,$$

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所以$\frac{\partial z}{\partial x}=0,\frac{\partial t}{\partial x}=0$,于是$\frac{\partial u}{\partial x}=f_1'$.

三个方程两边分别对 $y$ 求偏导,得

$$\begin{cases}\frac{\partial u}{\partial y}=f_2'+f_3'\frac{\partial z}{\partial y}+f_4'\frac{\partial t}{\partial y},\\ \frac{\partial g}{\partial y}+\frac{\partial g}{\partial z}\frac{\partial z}{\partial y}+\frac{\partial g}{\partial t}\frac{\partial t}{\partial y}=0,\\ \frac{\partial h}{\partial z}\frac{\partial z}{\partial y}+\frac{\partial h}{\partial t}\frac{\partial t}{\partial y}=0\end{cases}\Rightarrow \frac{\partial u}{\partial y}=f_2'+\frac{f_4'\frac{\partial g}{\partial y}\frac{\partial h}{\partial z}-f_3'\frac{\partial g}{\partial y}\frac{\partial h}{\partial t}}{\frac{\partial g}{\partial z}\frac{\partial h}{\partial t}-\frac{\partial g}{\partial t}\frac{\partial h}{\partial z}}.$$

**6. 解** $z=f(u)$ 两边对 $x,y$ 求偏导,得$\frac{\partial z}{\partial x}=f'(u)\frac{\partial u}{\partial x}$, $\frac{\partial z}{\partial y}=f'(u)\frac{\partial u}{\partial y}$.

方程 $u=\varphi(u)+\int_y^x P(t)\mathrm{d}t$ 两边对 $x,y$ 求偏导,得

$$\frac{\partial u}{\partial x}=\varphi'(u)\frac{\partial u}{\partial x}+P(x),\quad \frac{\partial u}{\partial y}=\varphi'(u)\frac{\partial u}{\partial y}-P(y),$$

解得$\frac{\partial u}{\partial x}=\frac{P(x)}{1-\varphi'(u)}$, $\frac{\partial u}{\partial y}=-\frac{P(y)}{1-\varphi'(u)}$,于是 $P(y)\frac{\partial z}{\partial x}+P(x)\frac{\partial z}{\partial y}=0$.

**7. 解** $\frac{\partial u}{\partial x}=\frac{1}{y}f_1'$, $\frac{\partial u}{\partial y}=-\frac{x}{y^2}f_1'+\frac{1}{z}f_2'$, $\frac{\partial u}{\partial z}=-\frac{y}{z^2}f_2'$;

$$\mathrm{d}u=\frac{1}{y}f_1'\mathrm{d}x+\left(\frac{1}{z}f_2'-\frac{x}{y^2}f_1'\right)\mathrm{d}y-\frac{y}{z^2}f_2'\mathrm{d}z;$$

$$\frac{\partial^2 u}{\partial y\partial z}=\frac{\partial}{\partial z}\left(\frac{\partial u}{\partial y}\right)=\frac{\partial}{\partial z}\left(-\frac{x}{y^2}f_1'+\frac{1}{z}f_2'\right)=-\frac{x}{y^2}\frac{\partial}{\partial z}(f_1')-\frac{1}{z^2}f_2'+\frac{1}{z}\frac{\partial}{\partial z}(f_2')$$

$$=\frac{x}{yz^2}f_{12}''-\frac{1}{z^2}f_2'-\frac{y}{z^3}f_{22}''.$$

**8. 解** $\mathrm{e}^{xy}-\ln(x^2+1)-y=0$ 两边同时对 $x$ 求导,得 $\mathrm{e}^{xy}\left(y+x\frac{\mathrm{d}y}{\mathrm{d}x}\right)-\frac{2x}{x^2+1}-\frac{\mathrm{d}y}{\mathrm{d}x}=0$,

解得

$$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\frac{2x}{x^2+1}-y\mathrm{e}^{xy}}{x\mathrm{e}^{xy}-1}=\frac{2x-(x^2+1)y\mathrm{e}^{xy}}{(x^2+1)(x\mathrm{e}^{xy}-1)};$$

$\mathrm{e}^z-\tan(2x+z^2)=0$ 两边同时对 $x$ 求导,得 $\mathrm{e}^z\frac{\mathrm{d}z}{\mathrm{d}x}-\sec^2(2x+z^2)\left(2+2z\frac{\mathrm{d}z}{\mathrm{d}x}\right)=0$,解得

$$\frac{\mathrm{d}z}{\mathrm{d}x}=\frac{2\sec^2(2x+z^2)}{\mathrm{e}^z-2z\sec^2(2x+z^2)},$$

则$\frac{\mathrm{d}u}{\mathrm{d}x}=f_1'+\frac{2x-(x^2+1)y\mathrm{e}^{xy}}{(x^2+1)(x\mathrm{e}^{xy}-1)}f_2'+\frac{2\sec^2(2x+z^2)}{\mathrm{e}^z-2z\sec^2(2x+z^2)}f_3'$.

**9. 解** (1) 因为 $0\leqslant|f(x,y)|\leqslant\sqrt{|xy|}$,所以$\lim\limits_{\substack{x\to0\\y\to0}}f(x,y)=0=f(0,0)$,故 $f(x,y)$ 在点 $(0,0)$ 处连续.

(2)$\Delta f(x,y)=f(x,y)-f(0,0)=\frac{\sqrt{|xy|}}{\rho^2}\sin\rho^2(\rho=\sqrt{x^2+y^2})$;

$f_x'(0,0)=\lim\limits_{x\to0}\frac{f(x,0)-f(0,0)}{x}=0$, $f_y'(0,0)=\lim\limits_{y\to0}\frac{f(0,y)-f(0,0)}{y}=0$.

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因为$\lim\limits_{\substack{\rho\to 0\\ y=x}}\dfrac{\Delta f(x,y)-f'_x(0,0)x-f'_y(0,0)y}{\rho}=\lim\limits_{\substack{\rho\to 0\\ y=x}}\dfrac{\sin\rho^2}{\rho^2}\cdot\dfrac{\sqrt{|xy|}}{\rho}=\dfrac{1}{\sqrt{2}}\neq 0$，所以$f(x,y)$在点$(0,0)$处不可微.

**10. 解** $0\leqslant|f(x,y)|\leqslant|xy|$.

因为$\lim\limits_{\substack{x\to 0\\ y\to 0}}|xy|=0$，由夹逼定理得$\lim\limits_{\substack{x\to 0\\ y\to 0}}f(x,y)=0=f(0,0)$，即$f(x,y)$在$(0,0)$处连续.

由$\lim\limits_{x\to 0}\dfrac{f(x,0)-f(0,0)}{x}=0$，得$f'_x(0,0)=0$，同理，$f'_y(0,0)=0$，即$f(x,y)$在$(0,0)$处可偏导.

令$\rho=\sqrt{x^2+y^2}$，$\dfrac{f(x,y)-f(0,0)-f'_x(0,0)x-f'_y(0,0)y}{\rho}=\dfrac{xy}{\rho}\sin\dfrac{1}{\rho^2}$；

$0\leqslant\left|\dfrac{xy}{\rho}\sin\dfrac{1}{\rho^2}\right|\leqslant|x|\dfrac{|y|}{\rho}\leqslant|x|$，由夹逼定理得$\lim\limits_{\rho\to 0}\dfrac{xy}{\rho}\sin\dfrac{1}{\rho^2}=0$，即$f(x,y)$在$(0,0)$处可微.

**11. 证明** 令$u=tx,v=ty,w=tz$，则$f(u,v,w)=f(tx,ty,tz)=t^kf(x,y,z)$，两边同时对$t$求导，得

$$x\frac{\partial f(u,v,w)}{\partial u}+y\frac{\partial f(u,v,w)}{\partial v}+z\frac{\partial f(u,v,w)}{\partial w}=kt^{k-1}f(x,y,z),$$

当$t=1$时，有$x\dfrac{\partial f}{\partial x}+y\dfrac{\partial f}{\partial y}+z\dfrac{\partial f}{\partial z}=kf(x,y,z)$.

**12. 证明** (1) 因为$\dfrac{\partial u}{\partial r}=\dfrac{\partial u}{\partial x}\cdot\dfrac{\partial x}{\partial r}+\dfrac{\partial u}{\partial y}\cdot\dfrac{\partial y}{\partial r}+\dfrac{\partial u}{\partial z}\cdot\dfrac{\partial z}{\partial r}=\dfrac{1}{r}\left(x\dfrac{\partial u}{\partial x}+y\dfrac{\partial u}{\partial y}+z\dfrac{\partial u}{\partial z}\right)=0$，所以$u$是不含$r$的函数，即$u$仅为$\theta$与$\varphi$的函数.

(2) 因为$\dfrac{\partial u}{\partial\theta}=\dfrac{\partial u}{\partial x}\cdot\dfrac{\partial x}{\partial\theta}+\dfrac{\partial u}{\partial y}\cdot\dfrac{\partial y}{\partial\theta}+\dfrac{\partial u}{\partial z}\cdot\dfrac{\partial z}{\partial\theta}=\dfrac{\partial u}{\partial x}\cdot(-r\sin\theta)\sin\varphi+\dfrac{\partial u}{\partial y}\cdot(r\cos\theta\sin\varphi)$

$$=-y\frac{\partial u}{\partial x}+x\frac{\partial u}{\partial y}=xy\left(-\frac{1}{x}\frac{\partial u}{\partial x}+\frac{1}{y}\frac{\partial u}{\partial y}\right)=0;$$

$$\frac{\partial u}{\partial\varphi}=\frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial\varphi}+\frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial\varphi}+\frac{\partial u}{\partial z}\cdot\frac{\partial z}{\partial\varphi}$$

$$=\frac{\partial u}{\partial x}\cdot(r\cos\theta\cos\varphi)+\frac{\partial u}{\partial y}\cdot(r\sin\theta\cos\varphi)+\frac{\partial u}{\partial z}\cdot(-r\sin\varphi).$$

令$\dfrac{1}{x}\dfrac{\partial u}{\partial x}=\dfrac{1}{y}\dfrac{\partial u}{\partial y}=\dfrac{1}{z}\dfrac{\partial u}{\partial z}=t$，则$\dfrac{\partial u}{\partial x}=tx,\dfrac{\partial u}{\partial y}=ty,\dfrac{\partial u}{\partial z}=tz$.

从而$\dfrac{\partial u}{\partial\varphi}=t(r^2\cos^2\theta\cos\varphi\sin\varphi)+t(r^2\sin^2\theta\cos\varphi\sin\varphi)+t(-r^2\sin\varphi\cos\varphi)=0$，故$u$仅是$r$的函数，即$u$不含$\theta$与$\varphi$.

**13. 证明** 由$\begin{cases}u=x,\\ v=\dfrac{1}{y}-\dfrac{1}{x},\end{cases}$得$\begin{cases}x=u,\\ y=\dfrac{u}{1+uv},\end{cases}$则

$$\frac{\partial\varphi}{\partial u}=-\frac{1}{z^2}\left(\frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial u}+\frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial u}\right)+\frac{1}{x^2}\cdot\frac{\partial x}{\partial u}=-\frac{1}{z^2}\left[\frac{\partial z}{\partial x}+\frac{1}{(1+uv)^2}\frac{\partial z}{\partial y}\right]+\frac{1}{u^2}$$

$$=-\frac{1}{z^2}\left(\frac{\partial z}{\partial x}+\frac{y^2}{x^2}\frac{\partial z}{\partial y}\right)+\frac{1}{u^2}=-\frac{1}{x^2}+\frac{1}{u^2}=0.$$

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**14. 解** 令 $\rho=\sqrt{(x-1)^2+y^2}$，由 $\lim\limits_{\substack{x\to1\\y\to0}}\dfrac{f(x,y)-2x-y-3}{(x-1)^2+y^2}=1$，得 $f(1,0)=5$ 且

$$f(x,y)-2x-y-3=o(\rho)\Rightarrow f(x,y)=2x+y+3+o(\rho).$$

由 $\Delta z=f(x,y)-f(1,0)=2(x-1)+y+o(\rho)$，得 $\mathrm{d}z\mid_{(1,0)}=2\mathrm{d}x+\mathrm{d}y$，故

$$f'_x(1,0)=2,f'_y(1,0)=1.$$

$$\lim_{h\to0}\frac{f(\mathrm{e}^{2h},h)-f[1+\ln(1+h),-h]}{h}=\lim_{h\to0}\frac{f(\mathrm{e}^{2h},h)-f[1+\ln(1+h),h]}{h}+\lim_{h\to0}\frac{f[1+\ln(1+h),h]-f[1+\ln(1+h),-h]}{h},$$

而 $f(\mathrm{e}^{2h},h)-f[1+\ln(1+h),h]=f'_1(\xi,h)[\mathrm{e}^{2h}-1-\ln(1+h)]$，$1+\ln(1+h)<\xi<\mathrm{e}^{2h}$；$f[1+\ln(1+h),h]-f[1+\ln(1+h),-h]=f'_2[1+\ln(1+h),\eta]\cdot2h$，$\eta$ 介于 $-h$ 与 $h$ 之间，因为 $f(x,y)$ 连续可偏导，所以

$$\lim_{h\to0}\frac{f(\mathrm{e}^{2h},h)-f[1+\ln(1+h),h]}{h}=\lim_{h\to0}f'_1(\xi,h)\cdot\frac{(\mathrm{e}^{2h}-1)-\ln(1+h)}{h}=f'_1(1,0)=2,$$

$$\lim_{h\to0}\frac{f[1+\ln(1+h),h]-f[1+\ln(1+h),-h]}{h}=2\lim_{h\to0}f'_2[1+\ln(1+h),\eta]=2f'_2(1,0)=2,$$

故原式 $=4$.

**15. 解**

$$\frac{\partial g}{\partial u}=\frac{\partial f}{\partial x}\cdot\frac{\partial x}{\partial u}+\frac{\partial f}{\partial y}\cdot\frac{\partial y}{\partial u}=v\frac{\partial f}{\partial x}+u\frac{\partial f}{\partial y},$$

$$\frac{\partial g}{\partial v}=\frac{\partial f}{\partial x}\cdot\frac{\partial x}{\partial v}+\frac{\partial f}{\partial y}\cdot\frac{\partial y}{\partial v}=u\frac{\partial f}{\partial x}-v\frac{\partial f}{\partial y},$$

$$a\left(\frac{\partial g}{\partial u}\right)^2-b\left(\frac{\partial g}{\partial v}\right)^2=a\left(v\frac{\partial f}{\partial x}+u\frac{\partial f}{\partial y}\right)^2-b\left(u\frac{\partial f}{\partial x}-v\frac{\partial f}{\partial y}\right)^2$$

$$=(av^2-bu^2)\left(\frac{\partial f}{\partial x}\right)^2+(au^2-bv^2)\left(\frac{\partial f}{\partial y}\right)^2+2uv(a+b)\frac{\partial f}{\partial x}\cdot\frac{\partial f}{\partial y}=u^2+v^2,$$

又 $\left(\dfrac{\partial f}{\partial x}\right)^2+\left(\dfrac{\partial f}{\partial y}\right)^2=4$，所以

$$(a+b)(v^2-u^2)\left(\frac{\partial f}{\partial x}\right)^2+2uv(a+b)\frac{\partial f}{\partial x}\cdot\frac{\partial f}{\partial y}+4(au^2-bv^2)=u^2+v^2,$$

于是 $\begin{cases}a+b=0,\\4a=1,\\4b=-1,\end{cases}$ 故 $a=\dfrac{1}{4},b=-\dfrac{1}{4}$.

**16. 解** 求 $f(x,y)$ 在区域 $D$ 的边界上的最值.

在 $L_1:y=0(0\leqslant x\leqslant6)$ 上，$z=0$；

在 $L_2:x=0(0\leqslant y\leqslant6)$ 上，$z=0$；

在 $L_3:y=6-x(0\leqslant x\leqslant6)$ 上，$z=-2x^2(6-x)=2x^3-12x^2$.

由 $\dfrac{\mathrm{d}z}{\mathrm{d}x}=6x^2-24x=0$，得 $x=4$，因为 $f(0,6)=0,f(6,0)=0,f(4,2)=-64$，所以 $f(x,y)$ 在 $L_3$ 上最小值为 $-64$，最大值为 $0$.

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在区域 $D$ 内，由 $\begin{cases}\dfrac{\partial z}{\partial x}=2xy(4-x-y)-x^2y=0,\\ \dfrac{\partial z}{\partial y}=x^2(4-x-y)-x^2y=0,\end{cases}$ 得驻点为(2,1).

$$A=\left.\frac{\partial^2 z}{\partial x^2}\right|_{(2,1)}=-6,B=\left.\frac{\partial^2 z}{\partial x\partial y}\right|_{(2,1)}=-4,C=\left.\frac{\partial^2 z}{\partial y^2}\right|_{(2,1)}=-8,$$

因为 $AC-B^2>0$ 且 $A<0$，所以(2,1)为 $f(x,y)$ 的极大值点，极大值为 $f(2,1)=4$.

故 $z=f(x,y)$ 在 $D$ 上的最小值为 $m=f(4,2)=-64$，最大值为 $M=f(2,1)=4$.

**17. 解** 当 $4x^2+y^2<25$ 时，由 $\begin{cases}z'_x=2x+12y=0,\\ z'_y=12x+4y=0,\end{cases}$ 得驻点为 $(x,y)=(0,0)$.

当 $4x^2+y^2=25$ 时，令 $F=x^2+12xy+2y^2+\lambda(4x^2+y^2-25)$.

由 $\begin{cases}F'_x=2x+12y+8\lambda x=0,\\ F'_y=12x+4y+2\lambda y=0,\\ F'_\lambda=4x^2+y^2-25=0,\end{cases}$ 得 $(x,y)=(\pm 2,\mp 3),\left(\pm\dfrac{3}{2},\pm 4\right)$.

因为 $z(0,0)=0,z(\pm 2,\mp 3)=-50,z\left(\pm\dfrac{3}{2},\pm 4\right)=106\dfrac{1}{4}$，所以目标函数的最大值和最小值分别为 $106\dfrac{1}{4}$ 和 $-50$.

**18. 解** 设 $P(x,y)$ 为椭圆 $x^2+4xy+5y^2=1$ 上的一点，原点到椭圆的距离为 $d=\sqrt{x^2+y^2}$.

令 $F=x^2+y^2+\lambda(x^2+4xy+5y^2-1)$.

由 $\begin{cases}F'_x=2x+\lambda(2x+4y)=0,\\ F'_y=2y+\lambda(4x+10y)=0,\\ F'_\lambda=x^2+4xy+5y^2-1=0,\end{cases}$ 得 $\begin{cases}(\lambda+1)x+2\lambda y=0,\\ 2\lambda x+(5\lambda+1)y=0,\\ x^2+4xy+5y^2-1=0.\end{cases}$

因为 $\begin{cases}x=0,\\ y=0\end{cases}$ 不是方程组的解，所以 $\begin{vmatrix}\lambda+1 & 2\lambda\\ 2\lambda & 5\lambda+1\end{vmatrix}=0$，解得

$$\lambda_1=-3-2\sqrt{2},\lambda_2=-3+2\sqrt{2}.$$

当 $\lambda_1=-3-2\sqrt{2}$ 时，$y=(1-\sqrt{2})x$，代入得 $x^2+y^2=\dfrac{2-\sqrt{2}}{10-7\sqrt{2}}$；

当 $\lambda_2=-3+2\sqrt{2}$ 时，$y=(\sqrt{2}+1)x$，代入得 $x^2+y^2=\dfrac{2+\sqrt{2}}{10+7\sqrt{2}}$.

则短半轴为 $d_{\min}=\sqrt{\dfrac{2+\sqrt{2}}{10+7\sqrt{2}}}$，长半轴为 $d_{\max}=\sqrt{\dfrac{2-\sqrt{2}}{10-7\sqrt{2}}}$，所求的椭圆面积为

$$S=\pi d_{\min}\cdot d_{\max}=\pi\sqrt{\frac{(2-\sqrt{2})(2+\sqrt{2})}{(10-7\sqrt{2})(10+7\sqrt{2})}}=\pi.$$

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